Project 1

Brandon Lim

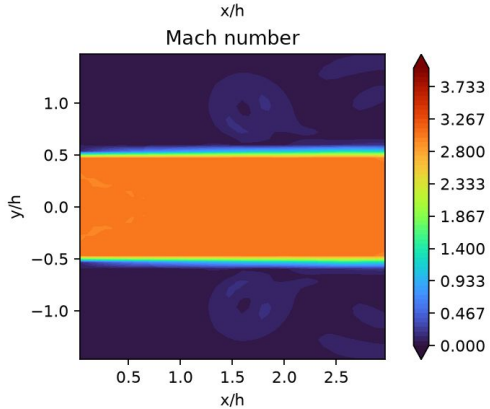
Problem 1

a) Code works correctly
b) There should be no shock wave because the pressures equal. This means that the flow is fully isentropic.

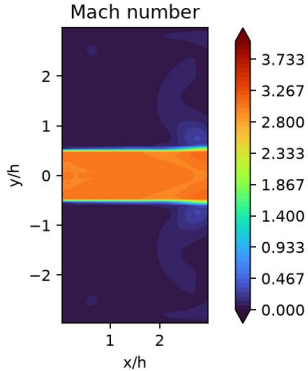
C)

The domain size determines the extent of the flow physics that can be captured and analyzed, while also influencing the spatial resolution for a fixed number of grid points. Boundary conditions define the behavior of the flow at the domain boundaries using known or assumed theoretical conditions. For this particular flow, the domain size influences the separation between the jet and the surrounding undisturbed flow. If the domain is too small, the boundaries may be positioned too close to the jet, where the flow is still significantly disturbed. This can introduce numerical artifacts and produce inaccurate flow predictions because the imposed boundary conditions may no longer represent the actual flow behavior at those locations. Consequently, the proximity of the boundaries to the jet can influence the computed flow field, particularly near the edges of the domain, and potentially distort the physical behavior of the jet itself.

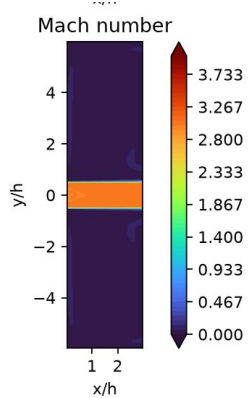
Ly/h = 1.5



Ly/h = 3

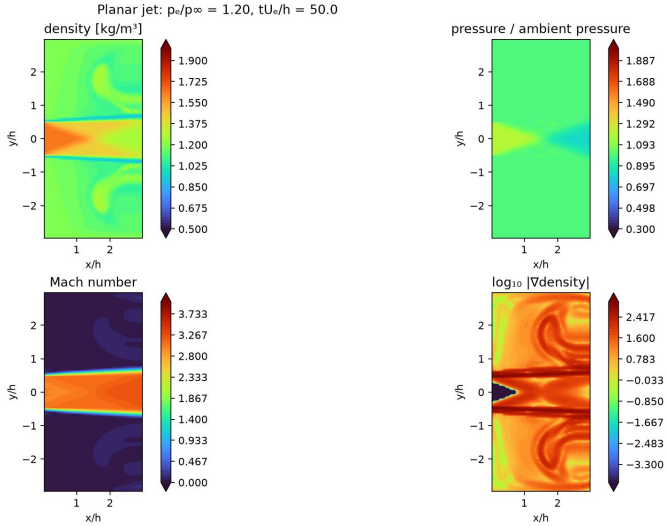


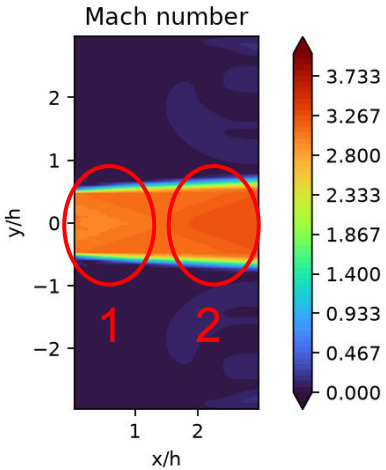
Ly/h = 6

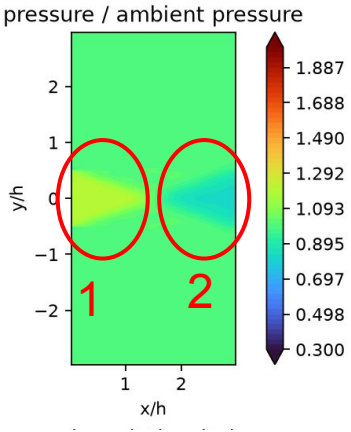


d)

Yes, we are capturing the expected behavior of an underexpanded nozzle. Since the nozzle exit pressure is greater than the ambient pressure, the flow initially undergoes an expansion that accelerates the fluid while decreasing its pressure. This behavior is evident in the region labeled State 1 in the plots below. As the flow continues downstream, it undergoes a series of expansions and compressions as it adjusts toward the ambient pressure, producing the characteristic shock diamond structure. The region labeled State 2 shows a further decrease in pressure and increase in velocity, indicating an additional expansion of the flow.


Planar jet: pₑ/p∞ = 1.20, tUₑ/h = 50.0
density [kg/m³]
1.900
1.725
1.550
1.375
1.200
1.025
0.850
0.675
0.500
pressure / ambient pressure
1.887
1.688
1.490
1.292
1.093
0.895
0.697
0.498
0.300
Mach number
3.733
3.267
2.800
2.333
1.867
1.400
0.933
0.467
0.000
log₁₀ |∇density|
2.417
1.600
0.783
−0.033
−0.850
−1.667
−2.483
−3.300
y/h
x/h

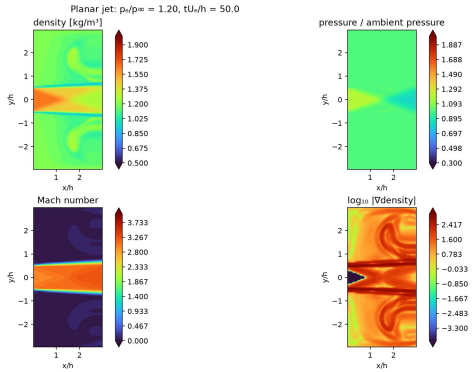

Mach number
3.733
3.267
2.800
2.333
1.867
1.400
0.933
0.467
0.000
1
2
y/h
x/h


pressure / ambient pressure
1.887
1.688
1.490
1.292
1.093
0.895
0.697
0.498
0.300
1
2
y/h
x/h

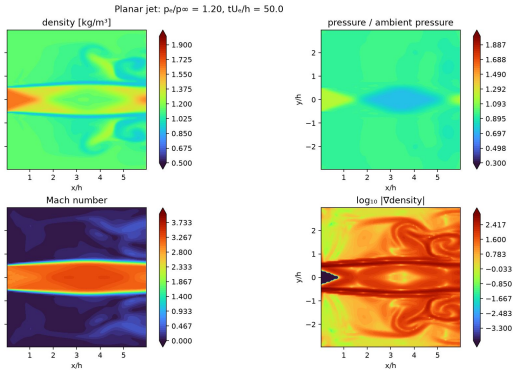
e)

The domain size determines the extent of the flow physics that can be captured and analyzed, while also influencing the spatial resolution for a fixed number of grid points. Boundary conditions define the flow behavior at the domain boundaries based on known or assumed theoretical conditions. For this particular flow, the domain size influences the separation between the jet and the surrounding undisturbed flow. If the domain is too small, the boundaries may be positioned too close to the jet, where the flow is still significantly disturbed. This can introduce numerical artifacts and inaccurate flow predictions because the imposed boundary conditions may not accurately represent the physical conditions at those locations. Consequently, the proximity of the boundaries to the jet can affect the computed flow field, particularly near the domain edges, and potentially distort the jet's behavior. By increasing the domain length in the x-direction, we can capture more of the downstream flow development and observe the propagation of the shock diamond structure more completely. However, for a fixed number of grid points, increasing the domain length also increases the grid spacing, potentially reducing the resolution of the shock structures.

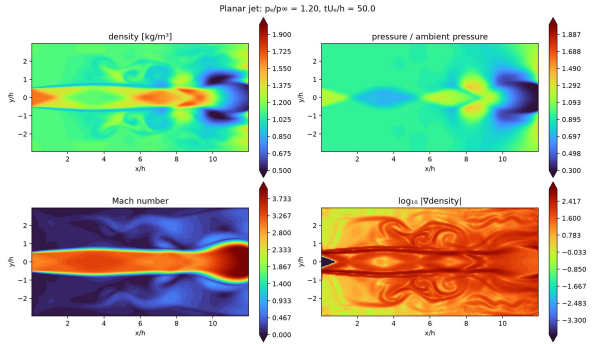
Lx/h = 3



Lx/h = 6

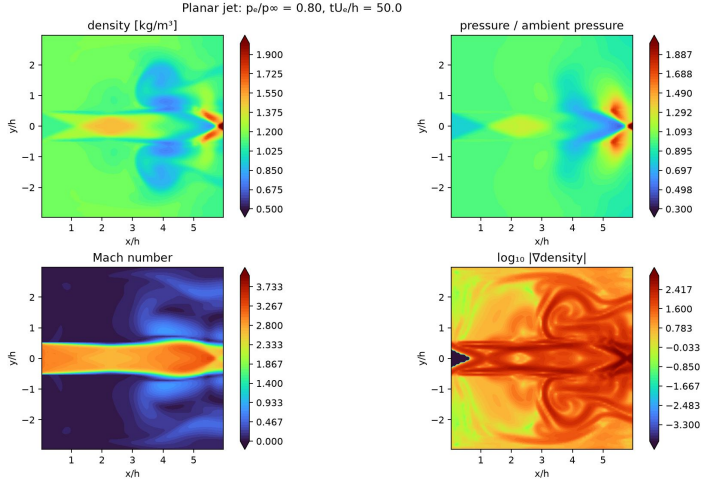


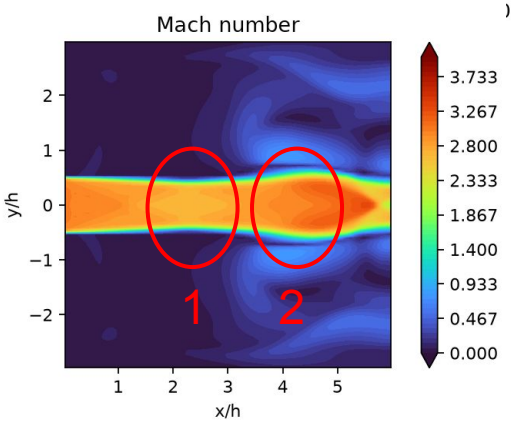
Lx/h = 9

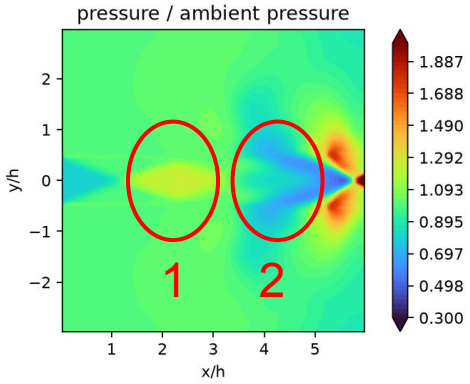


f)

Yes, the simulation is capturing the expected behavior of an overexpanded nozzle. Since the nozzle exit pressure is lower than the ambient pressure, shock waves form to compress the flow and increase its pressure toward ambient conditions. This behavior is evident in the region labeled State 1 in the plots below, where the pressure increases across the shock while the velocity decreases. In State 2, the flow pressure has increased beyond the ambient pressure due to the initial compression, resulting in an expansion that decreases the pressure and accelerates the flow. This alternating process of compression and expansion continues downstream, producing the characteristic shock diamond structure observed in the simulation.


Planar jet: pₑ/p∞ = 0.80, tUₑ/h = 50.0
density [kg/m³]
pressure / ambient pressure
Mach number
log₁₀ |∇density|
y/h
x/h

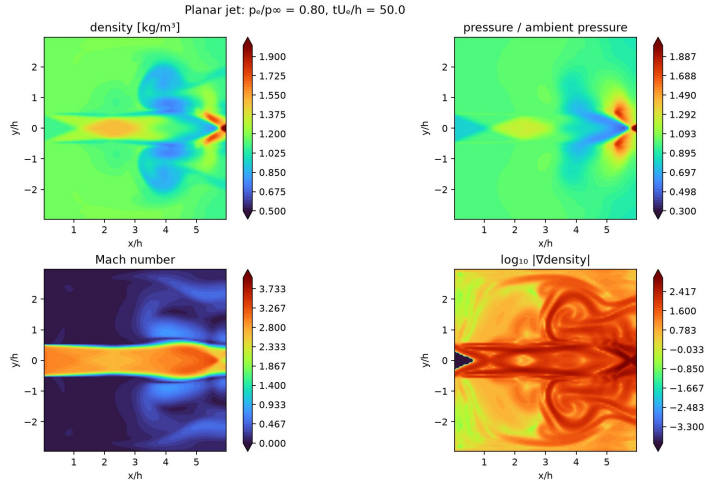

Mach number
1
2
y/h
x/h


pressure / ambient pressure
1
2
y/h
x/h

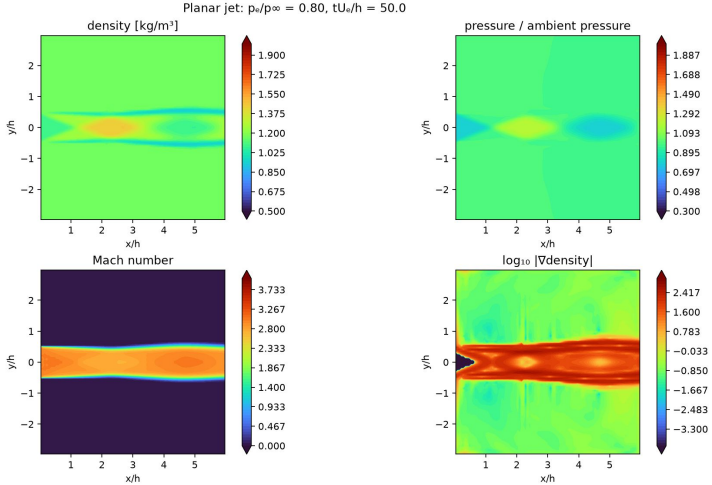
g)

Initializing the initial jet to 1 reduced the time it took to reach quasi-steady state but didn’t capture the fine details of the flow physics of the transient solution effects the steady state solution. Overall, the initial jet of 1 found the correct macro information and ran faster, making it useful for low computing power and cost while still studying major physical phenomena

Steady State Time for Initial_Jet = 0: ~45s

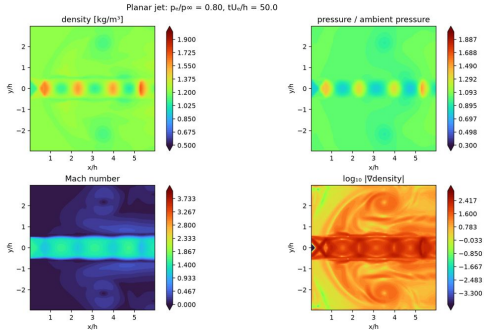


Steady State Time for Initial_Jet 1: ~20s

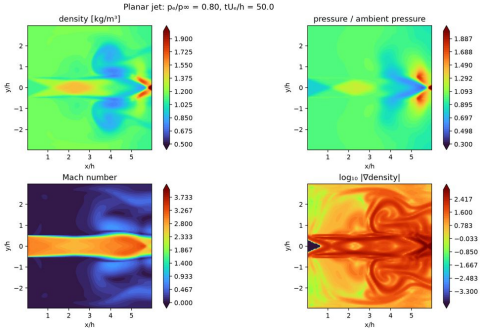


h)

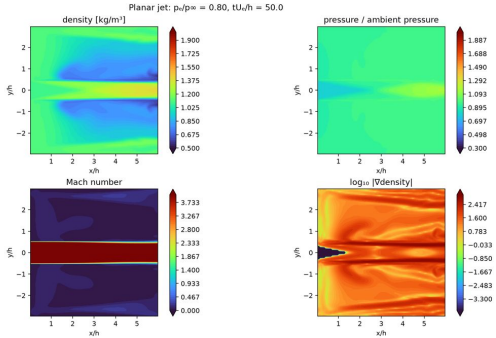
M = 1.5
Diamond x/h ~ 0.5

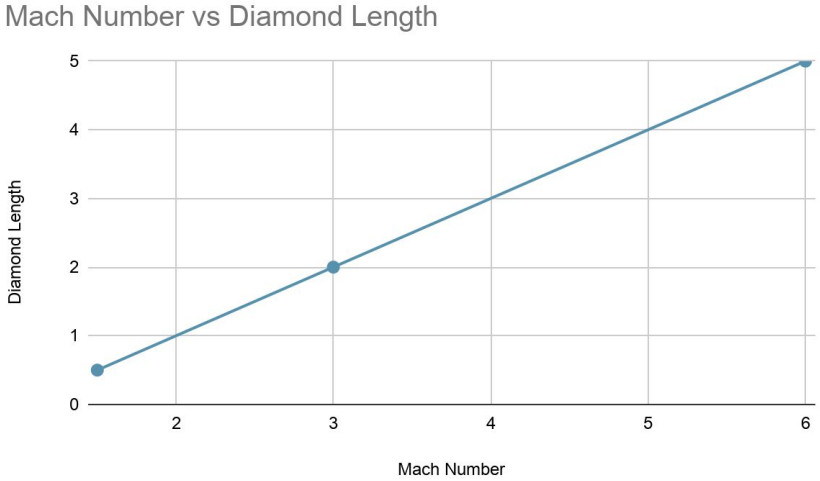


M = 3.0
Diamond x/h ~ 2



M = 6.0
Diamond x/h ~ 5



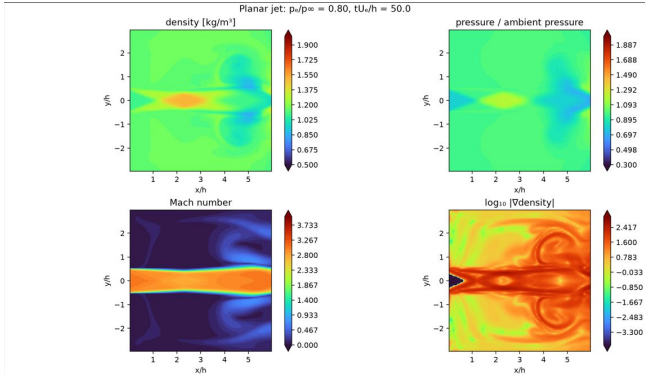


As the mach number gets higher, the shock diamond length seems to be getting longer at a linear rate.

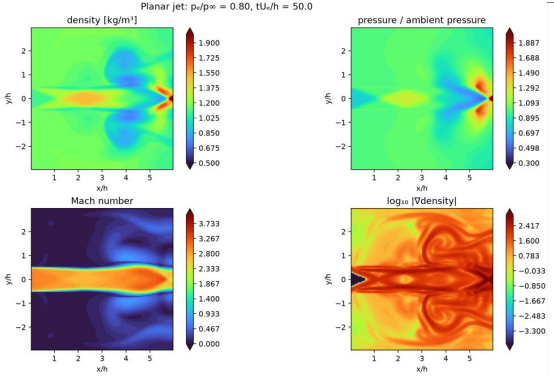
i)

With increasing gamma, the recirculating flow tends to increase in strength and size. Also, while looking at the pressure and density plots, we can see that when the ratio of specific heats increases, the shock diamond shape, locations, and intensity seem to be changing as well.

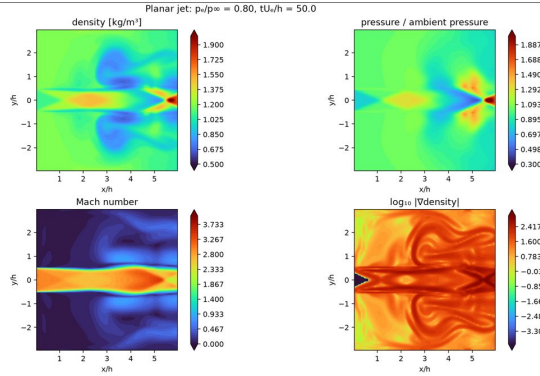
Gamma = 1.2



Gamma = 1.4



Gamma = 1.6



Problem 2

a)

$$\frac{p_2}{p_1} = \frac{2\gamma M_1^2 \sin^2 \beta}{\gamma+1} - \frac{\gamma-1}{\gamma+1}$$

- p2/p1 = 1/0.8
- Gamma = 1.4
- M1 = 3.0

Solving for beta:

Beta = 21.55 degrees

b)

Using the CSV Data, I was able to identify the column that corresponded to the end of the shock location by reviewing the pressure data and seeing where there was a big jump in pressure. I also estimated the row height of half of the shock from this CSV data. Using these number indicidies of rows and columns, I could calculate the slope of the shock line and therefore the shock angle.

$$\tan^{-1}\left(\frac{6}{13}\right)$$

= 24.78 degrees

My simulated angle is about 3.2 degrees bigger than my predicted angle.

c)

	Analytical	Simulated	%error
Static Temperature	277.2 Kelvin	275.3 Kelvin	0.69%
Static Density	1.27 kg/m^3	1.25 kg/m^3	1.57%
Mach Number	2.85	2.87	0.7%

d)

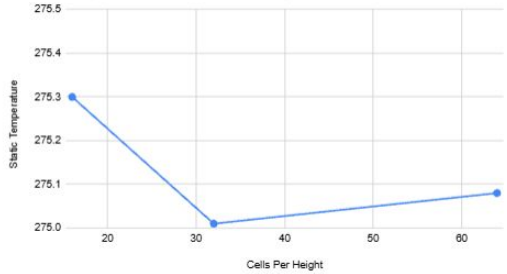
Looking at the csv data for a specific parameter, it takes about 3-4 rows for a quantity to change across a shock wave to where the values above these rows are homogenous and the values below these rows are homogeneous. With the current discretization scheme, that corresponds to a thickness of about y/h ~ 0.1875 where the theoretical shock thickness is infinitesimally small.

e)

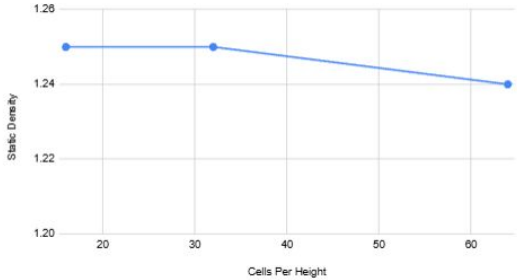
Overall, the shock states are somewhat consistent over grid resolution which makes sense because we're looking at averaged steady state behavior after a shock which does not change drastically under these resolution comparisons whereas their small scale fluctuations and smaller dynamics will show more detail as resolution increases. The shock angle and shock thickness become more accurate as grid resolution increases. The shock angle approaches our theoretically calculated one and the shock thickness continues to decrease in size towards zero.

	16 cells per height	32 cells per height	64 cells per height
Static Temperature	275.3 Kelvin	275.01 Kelvin	275.08 Kelvin
Static Density	1.25 kg/m^3	1.25 kg/m^3	1.24 kg/m^3
Mach Number	2.87	2.85	2.85
Shock Angle	24.78 degrees	21.66 degrees	21.47 degrees
Shock Thickness	y/h ~ 0.1875	y/h ~ 0.125	y/h ~ 0.09375

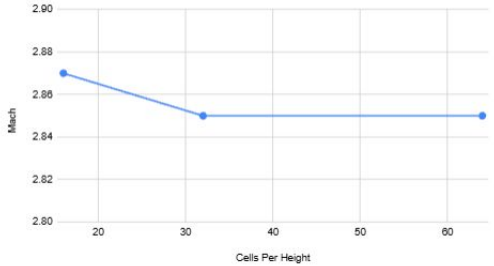
Static Temperature vs Cells Per Height



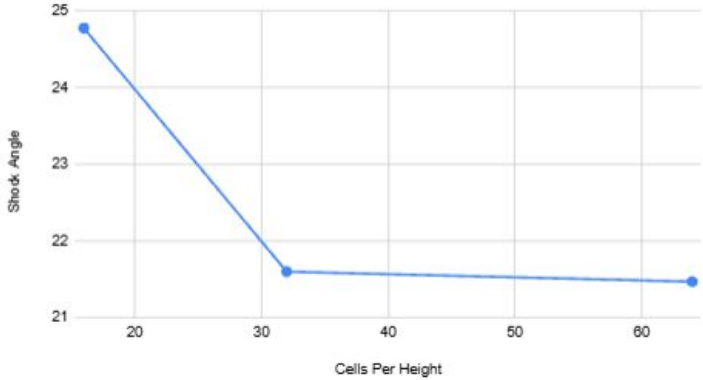
Static Density vs Cells Per Height



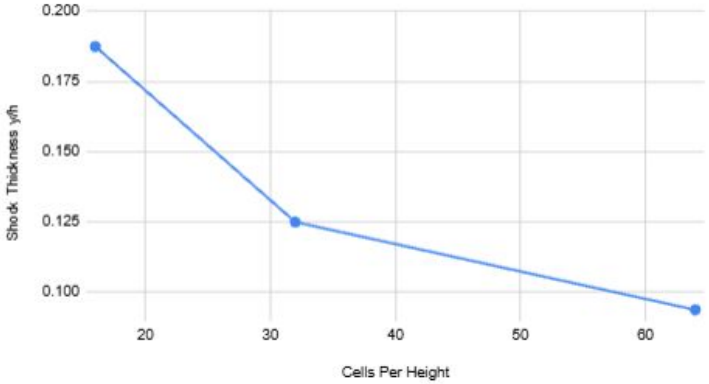
Mach vs Cells Per Height



Shock Angle vs Cells Per Height



Shock Thickness vs Cells Per Height



f)

The calculated theta angle from the first shock was 2.9 degrees. This should be the same angle to turn the flow back after the second shock. Knowing this, I can solve the the shock angle beta with the mach number and flow angle.

	State 2	State 3
Static Temperature	277.2 Kelvin	276.19 Kelvin
Static Density	1.27 kg/m^3	1.48 kg/m^3
Mach Number	2.85	2.71

g)

	State 3 Analytical	Simulated	%error
Static Temperature	276.19 Kelvin	276.5 Kelvin	0.11%
Static Density	1.48 kg/m^3	1.47 kg/m^3	0.68%
Mach Number	2.71	2.72	0.34%

These quantities match much better than the first shock. This is due to the fact that our first shock simulation used a cell height resolution of 16 while this simulation for the second shock used a cell height of 64. This gave much better accuracy in the state solutions.

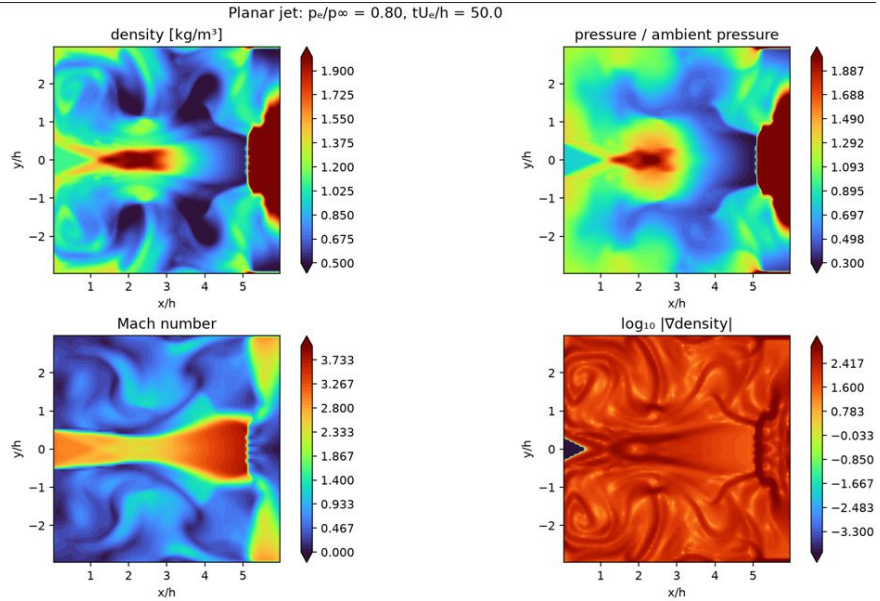
Problem 3

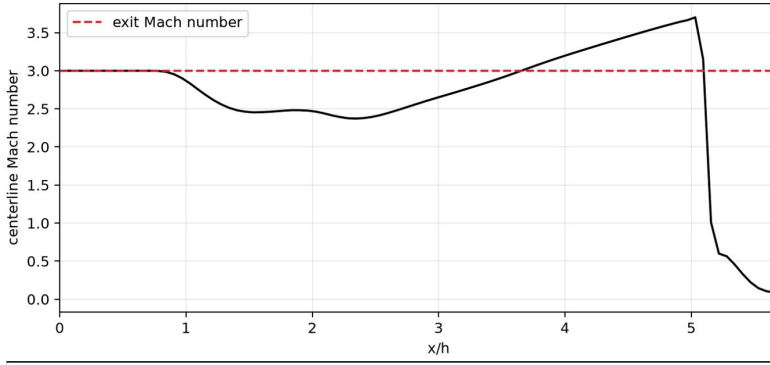
1)

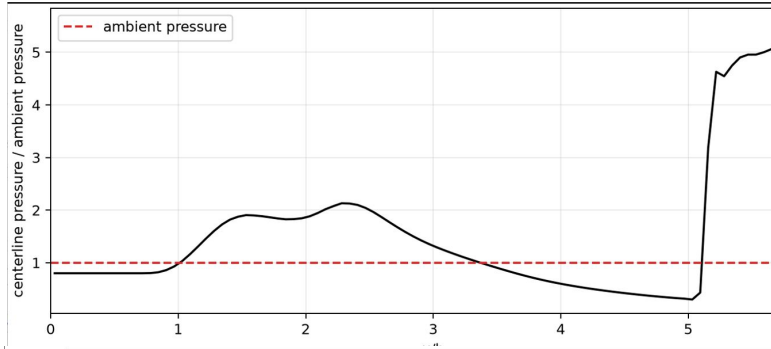
I changed the right boundary condition from a neuman to a inviscid slip wall to replicate having flow from a rocket hitting the floor of a launch pad.

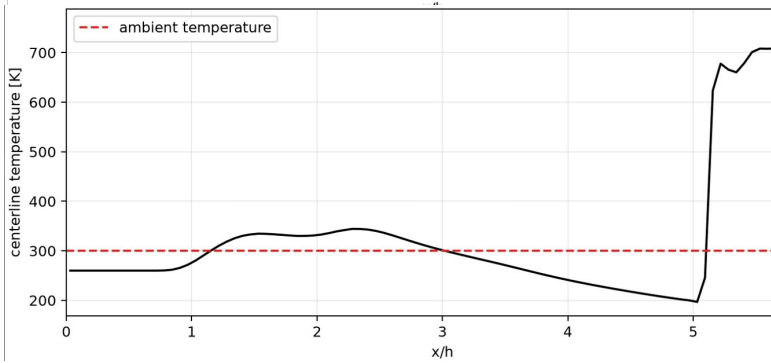
Mathematically, the value of the velocity normal to the right wall is equal to the negative of itself. This creates a no penetration condition where the wall is. This was similar to the approach taken in assignment two to classify the boundary conditions of the box.

2)


Planar jet: pe/p∞ = 0.80, tUe/h = 50.0
density [kg/m³]
pressure / ambient pressure
Mach number
log10 |∇density|
x/h
y/h

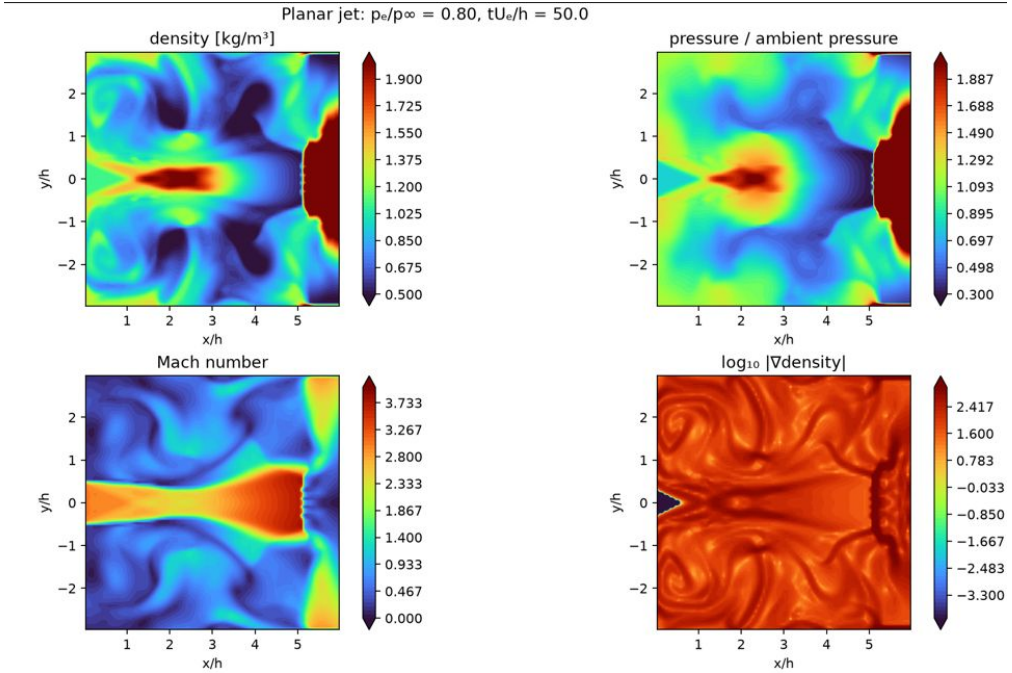

exit Mach number
centerline Mach number
x/h


ambient pressure
centerline pressure / ambient pressure


ambient temperature
centerline temperature [K]
x/h

2)

From the contour plots, we can see that when the jet approaches the wall, there is a huge change in pressure and velocity at about x/h = 5. This is due to the fact that a normal shock appears in this location which is what we would expect because the flow stagnates at the wall, requiring a strong normal shock to increase the pressure and drop the velocity.



3)

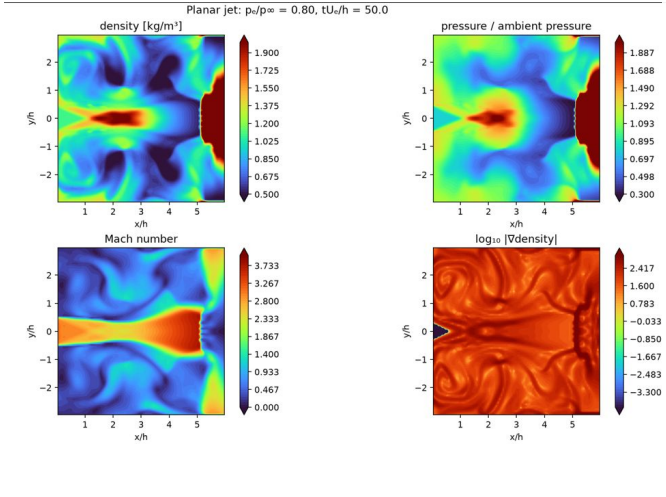
Using the inlet conditions as a pre normal shock states to simplify the analysis, I used the normal shock equations to estimate the states after the shock. This created some error because the states right before the normal shock were still being altered by the oblique shocks in the simulation.

	Analytical Post Normal Shock	Simulated Post Normal Shock	% error
Static Temperature	696.584 Kelvin	702 Kelvin	2.25%
Static Density	4.19 kg/m^3	5.74 kg/m^3	36.9%
Pressure	837620 Pa	1160000 Pa	38.4%

4)

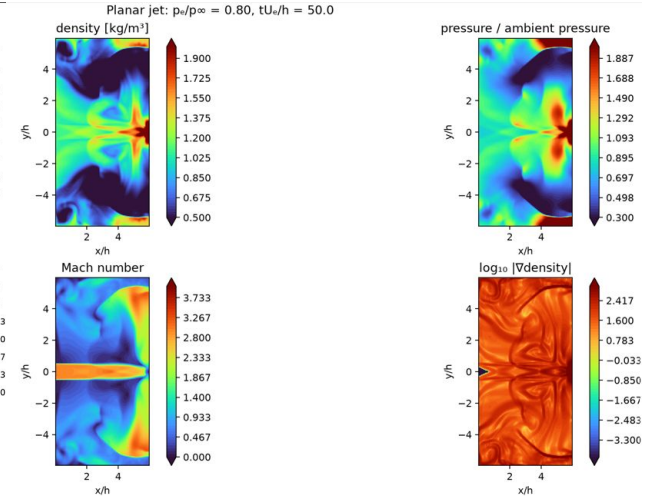
With the current convective time and y domain length, there is a lot of mixing flow as a result of the jet hitting the floor and creating recirculating flows that interact with each other. This makes it hard to study a steady state behavior. I’m going to change the y domain length to allow the recirculating vortices to diffuse outward rather than mix back into the jet and allow for enough time for this flow to become steady.

4)



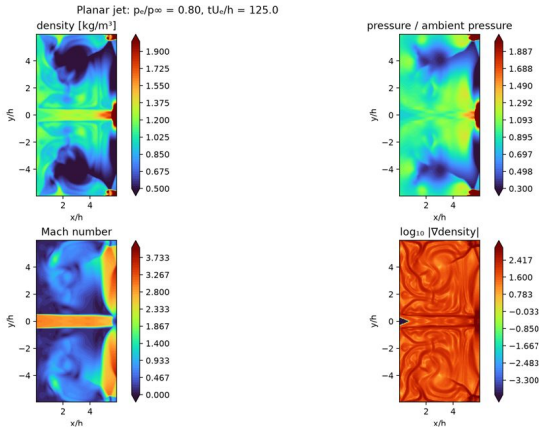
y/h = 6

T = 50 sec



y/h = 12

T = 50 sec



y/h = 12

T = 125 sec